

Abhinav Reddy Mettu

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SUMMARY

Aspiring Data Scientist with strong proficiency in Python and SQL for data analysis, modeling, and machine learning. Experienced in building end-to-end ML pipelines, including feature engineering, model tuning, deployment, and performance optimization on cloud platforms such as AWS. Hands-on experience with forecasting, deep learning frameworks (TensorFlow, PyTorch). Skilled in communicating insights using data visualization tools like Tableau, Power BI, Matplotlib, and Seaborn. Adept at applying statistics, probability, and hypothesis testing to solve complex business problems and support decision-making.

EDUCATION

The University of Texas at Dallas

Master's in Business Analytics and Artificial Intelligence (Concentration: **Data Science**)
Graduate Certificate in **Applied Machine Learning**

Dallas, TX

May 2025

Dec 2024

Jawaharlal Nehru Technological University

Bachelor of Technology - **Mechanical Engineering**

Hyderabad, India

June 2023

PROJECTS

Strategic Customer Retention through Predictive Churn Analysis in Telecom

February 2025 - April 2025

- Developed a predictive churn model using Logistic Regression, Decision Tree, and Random Forest, achieving 85% accuracy to help reduce customer attrition.
- Engineered churn-indicative features and conducted EDA, enabling retention strategy optimization and reducing potential churn by 15%.
- Applied evaluation metrics (confusion matrix, precision, recall) and optimized model performance using cross-validation and threshold tuning.
- Built and deployed a full ML pipeline using Python, reducing data processing time by 30% and improving operational efficiency.
- Visualized churn correlations using heatmaps, bar plots, and customer demographics to identify behavioral trends.
- Addressed data imbalance using resampling techniques and tested impact of SMOTE vs. class weights.
- Created customer risk segmentation dashboard in Tableau, providing business teams with real-time churn alerts and retention targeting tools.

Credit Card Fraud Detection Using Machine Learning

September 2024 - October 2024

- Developed a binary classification model using Logistic Regression, Random Forest, and XGBoost to detect fraudulent credit card transactions on an imbalanced dataset.
- Applied SMOTE and random undersampling techniques to address class imbalance, improving minority class representation and model reliability.
- Achieved 94% precision and 0.98 ROC-AUC on the minority class, significantly reducing false positives and minimizing fraud-related costs.
- Used SHAP values and feature importance plots in Python (Matplotlib, SHAP) to improve model interpretability and stakeholder transparency.
- Conducted PCA (Principal Component Analysis) to reduce dimensionality and improve model efficiency while retaining 95% variance.
- Implemented a pipeline to automate preprocessing, scaling, and model training, ensuring reproducibility and modular experimentation.
- Compared model performance across metrics including F1-score, precision-recall curve, and AUC using cross-validation and statistical testing.

Employee Attrition Prediction

May 2024 - July 2024

- Built classification models (Logistic Regression, Random Forest, XGBoost) on IBM's HR dataset to predict employee attrition.
- Conducted detailed EDA to identify key attrition drivers such as job role, overtime, and job satisfaction.
- Applied SMOTE to balance class distribution and optimized hyperparameters using GridSearchCV and Stratified K-Fold CV.
- Achieved 89% accuracy and 0.92 ROC-AUC, enabling HR to proactively address employee churn.
- Created an interactive Tableau dashboard to visualize attrition risk across departments and job levels.
- Performed feature transformation and correlation analysis to remove multicollinearity and improve model generalization.
- Implemented ensemble stacking using Logistic Regression as a meta-learner to combine base model outputs, boosting predictive accuracy.
- Delivered business insights through stakeholder-ready reports that identified high-risk profiles by department, performance rating, and overtime status.

Diabetes Risk Prediction

May 2022 - June 2022

- A healthcare provider aimed to proactively identify individuals at risk of developing Type 2 Diabetes.
- Built logistic regression and random forest models on health indicators (BMI, glucose, family history) to assess diabetes risk.
- Cleaned health datasets and conducted feature engineering using Python (Pandas, Scikit-learn); handled imbalance via SMOTE and optimized models with GridSearchCV.
- Achieved 87% accuracy and 82% recall, enabling earlier interventions and enhancing preventive care outcomes.
- Conducted univariate and bivariate statistical analysis to evaluate the impact of variables like glucose, insulin, and age on diabetes risk.
- Visualized model outputs and health trends using Seaborn and Matplotlib to communicate risk factors to non-technical stakeholders.
- Tuned model thresholds based on precision-recall tradeoffs to improve identification of high-risk patients without inflating false positives.

LEADERSHIP AND TRAINING EXPERIENCE

Assistant Coordinator – Training & Development, Advanced Academic Center, GRIET

May 2020 - May 2023

- Mentored 30+ students in machine learning and career planning, resulting in a 20% increase in student participation in research projects.
- Supervised a 4-member team in executing and presenting ML projects, promoting applied learning and analytical thinking.
- Coordinated college-wide project expos, increasing student engagement in innovation and research by 30%.
- Designed ML training strategies and built a collaborative peer-learning platform to develop foundational data science skills.

SKILLS

- Data Visualization & Reporting:** Tableau, Power BI, Excel (Pivot Tables, VLOOKUP, Dashboarding, Charts)
- Data Analysis & Querying:** SQL (MySQL, PostgreSQL), Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, XGBoost)
- Machine Learning & Analytics:** Forecasting, Classification, Regression, Feature Engineering, Hyperparameter Tuning (GridSearchCV), SMOTE
- Data Engineering Basics:** Data Cleaning, Data Wrangling, ETL Basics, Model Deployment
- Statistics & Analytics:** Hypothesis Testing, ANOVA, EDA, Predictive Modeling, Business Intelligence
- Certifications:** AWS AI Practitioner [View Certificate](#)

RESEARCH PAPER - ICMED-ICMPC 2023, International Conference on Design and Manufacturing Aspects for Sustainable Energy- May 2023

Author – [Optimisation and Mechanical Characterization of Casein and Seaweed Resin with Hemp Reinforcement](#)